

Project Design Phase

Solution Architecture

Date	14-07-2026
Project Name	Customer Registry

Solution Architecture:

The Customer Complaint Management System is developed using the MERN Stack architecture. The system provides role-based access for Customers, Administrators, and Agents. Customers can register and submit complaints through the web interface. Administrators manage users and assign complaints to agents. Agents update complaint status and resolve issues. The application ensures secure authentication, centralized complaint management, and real-time status tracking.

The architecture consists of:

- Frontend developed using React.js for user interaction.
- Backend developed using Node.js and Express.js to handle business logic and API requests.
- MongoDB database for storing users, complaints, assignments, and status information.
- JWT Authentication for secure login and role-based authorization.
- REST APIs for communication between frontend and backend.
- Cloud/File Storage for storing uploaded complaint-related files if required.
- Dashboard and Reporting module for complaint monitoring and analytics.

System Flow:

1. Customer registers and logs into the system.
2. Customer submits a complaint.
3. Complaint details are stored in MongoDB.
4. Admin views and assigns complaints to agents.
5. Agent updates complaint status (Pending, In Progress, Resolved).
6. Customer tracks complaint status in real time.
7. Admin monitors reports and overall complaint resolution progress.

Architecture

